

National Productivity and Quality Specifications (NPQS)

A1-50 Architectural Precast Concrete Cladding

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1. GENERAL

Read with the General Requirements section, and all other contract documents.

1.1 Scope

This section covers the requirements for architectural aspects of precast concrete cladding including elements such as façade panels, screens and sunbreakers. It covers mainly the finishes aspect of the precast elements. Requirements for concrete and reinforcement are set out in the structural specifications.

Use in combination with section A1-20 when precast units incorporate windows installed at the precast yard. Use in combination with section A1-60 when precast units incorporate stone claddings.

1.2 Related Sections

Read this work section in conjunction with the relevant requirements of the other work sections including the following:

A1-20 Windows

A1-60 Stone Cladding

1.3 Standards

Unless otherwise agreed by the SO, ensure all of the Works comply with the relevant requirements of the Standards and Codes listed below or referenced in the body of the Specification. Alternative Standards and Codes may be proposed for approval by the SO, provided it can be demonstrated that the alternative Standards and Codes comply with the requirements of the standards specified. All Standards and Codes quoted are the current version, unless specific year references are noted.

Singapore standards

SS CP33	Lightning protection
SS CP81	Precast concrete slab and wall panels
SS CP65	Structural use of concrete
SS 26	Ordinary Portland cement
SS 31	Aggregates
SS 32	Steel fabric for reinforcement of concrete
SS 289	Specification for Concrete
SS 320	Admixtures
SS 18	Cold reduced steel wire for reinforcement of concrete
BS 1014	Pigments for Portland cement
BS 4255	Specification for non-cellular gaskets
BS 5080 parts 1&2	Structural fixings in concrete
BS 5250 and CIBSE handbook	Code of practice for control of condensation in buildings
BS 5889	Silicone sealants
BS 5950	Structural use of steel
BS 6213	Selection of Construction Sealants
BS 6213	Loading of buildings
BS 6744	Stainless steel bars for concrete reinforcement
BS7079	Steel preparation

BS EN 24032, 24033	Carbon steel bolts and nuts
BS EN 10056 – 1&2	Carbon steel angles
BS EN ISO 11600	Building construction. Jointing products. Classification and requirements for sealants.
BS EN 100025	Hot rolled steel
BS EN 10029	Stainless steel
BS EN 10048	Stainless steel
BS EN 10088	Stainless steel
BS EN 10258	Cold-rolled stainless steel and narrow strip and cut lengths – tolerances on dimensions and shape
BS EN 10259	Cold-rolled stainless steel wide strip and plate/steel – tolerances on dimensions and shape
BS EN 10021,	General technical delivery requirements for steel and iron products
BS EN ISO 3506 parts 1&2	Stainless steel fasteners
BS EN 1461	Galvanising
PD 6484	Bimetallic corrosion
CIBSE handbook	Thermal performance

1.4 Trade Preamble

1.4.1 Contractor's Submissions and Proposals

The design intent including type and locations of the precast cladding systems for the project are indicated in the drawings and schedules. Base on the information supplied, engage qualified and experienced personnel to carry out and submit the following items for the SO's acceptance:

1.4.1.1 Design Development

Develop all necessary details for the fabrication and installation of the cladding system(s), and to meet with the performance requirements as set out in clause 2.1 below. Details to develop may include shape and sizing of components, structural support, accessories, joints, waterproofing and connections to the building etc.

1.4.1.2 Mould and formwork

Design all mould and formwork for the fabrication of the precast elements including all accessories for the lifting, handling and storage of the elements.

1.4.1.3 Temporary supports

Design to provide all necessary temporary supports and ensure that the precast elements are stable after erection but before the building structure is completed. Provide all necessary precautionary measures to ensure that the structure is not subjected to loads that will cause short or long-term distress to any part of the concrete structures.

1.4.1.4 Manufacturers and Products

Propose manufacturers and products for components of the concrete cladding systems when called for.

1.4.1.5 Methods

Propose method of fabrication, erection/installation.

1.4.1.6 Weather/waterproofing

Propose details of weather/waterproofing.

1.4.2 Shop Drawings

Produce shop drawings to enable the fabrication and installation of the cladding system.

1.4.3 Co-ordination with other Works

Co-ordinate the precast concrete cladding works particularly the interfacing with the following trades:

- a. External and internal masonry
- b. Structural frame
- c. Windows and doors (factory fitted or site installed into openings)
- d. Lightning protection systems
- e. External finishes
- f. Mechanical and electrical services

1.4.4 Provide Spare Materials

Deliver to site in strong protective packages marked for identification, and store where directed, components and materials for future replacement and repair. Refer to PSD for list.

1.4.5 Submission to Authorities

Engage a PE (Structural) to carry out the design of the structural supports for the cladding system and submit for BCA approval as a QP for this design.

1.4.6 Warranty

Provide a ten-year warranty in accordance with contract conditions for the entire cladding system including all components and accessories for the specified performance.

1.4.7 Maintenance Manual

Refer to the PSD for any requirement to provide a maintenance manual.

1.4.8 Quality Control Plan

Refer to the PSD for any requirement to provide a quality control plan.

2. PERFORMANCE REQUIREMENTS

2.1 Contractor's Brief

When carrying out the proposals as set out in clause 1.4.1, take account of the following requirements:

2.1.1 General Requirement

Precast concrete cladding systems shall (be / have):

- a. Secondary drainage system behind the weathering seals, that drains to the outside
- b. Concrete cover to all reinforcement shall comply with SS CP 65, subject to a minimum reinforcement cover 30mm unless otherwise agreed.
- c. No spacers visible or less than 10mm sub-surface in the finished panel
- d. No voids or honeycombing
- e. Internal air seal and vapour barrier continuous past slab edge
- f. No standing water

2.1.2 Resistance to Climatic and Environmental Conditions

2.1.2.1 Weather resistance

Limit airflow through the precast concrete cladding, from the exterior surface to the interior surface through all joints, to the specified air leakage rates quoted below when subjected to wind induced pressures or suctions across the cladding of 600 Pa] to not more than 1.5 cu m/hr per metre length of joint. There shall be no region of concentrated airflow through the precast concrete cladding joints or at any interfaces.

Prevent leakage of water into the interior of the building from the outer face of the assembly, resulting from wind pressure kinetic energy, gravity, surface tension, or capillary action with any combination of wind and precipitation likely to be experienced, up to and including 600 Pa static pressure for any period of 5 minutes, and 900 Pa dynamic pressure, for 3 second periods at random intervals.

Ventilated cavities to be compartmentalised by the use of flashings and baffles. No cavity shall be more than 12m high and 50m long. Vertical flashings or baffles to be added at corners.

Prevent water entering into those parts of the precast concrete cladding that would be adversely affected by the presence of water.

Maintain weathertightness under reasonably anticipated imposed loads and thermal, structural or other movements.

No seepage to occur through the concrete.

2.1.2.2 Avoidance of algae growth

Where applicable, agree suitable measures with the SO for protecting against algae growth on the surface of the cladding.

Sealants must be tested on sample panels and approved by the SO.

2.1.2.3 Thermal Stress

Resist cracking to stone, glass ceramic materials caused by thermal stress.

2.1.2.4 Condensation performance

Resist the formation of condensation on visible surfaces, interstices or locations where it is unable to drain.

2.1.2.5 Resistance to acid rain

Obtain local pollution data and rainfall data.

Design the precast concrete cladding to maintain integrity and appearance for the design life of the building

Advise any coatings, maintenance and cleaning treatments necessary.

2.1.2.6 Design for thermal and condensation performance

Use insulation of appropriate performance to provide adequate level of performance without localised cold bridging where insulation must be locally reduced in thickness such as anchor positions.

Use vapour barriers as necessary to eliminate condensation around the cladding.

2.1.2.7 Climatic Data

Assume:

Ambient temperature range	16°C to 40°C
Radiant temperature range	16°C to 80°C
Relative humidity range	90%±10%
Environment	Coastal

Gather any additional climatic data required from the Singapore Meteorological Observatory.

Submit for review to the SO any additional assumptions concerning temperatures, humidity levels, etc.

2.1.3 Structural Design and Loadings

2.1.3.1 General

Refer to structural specifications on precast concrete works.

2.1.3.2 Vibration and repetitive loading

Resist working loose of anchors and other components due to the effects of vibrations, or cyclic effects of load, deflections and thermal expansion.

2.1.3.3 Locked in stresses

Avoid the introduction of locked-in stresses that may be detrimental to the performance of the precast cladding.

2.1.4 Building Movements and Structural Tolerance

2.1.4.1 Structural Movement and Movement Between Works by Others

Accommodate the following movements in combination; without any reduction in the specified performance, refer to the SO for further guidance and data for calculation input:

- a. Deflections and movements of the structure and building wall under design loads.
- b. Changes in dimension and shape of building elements arising from building movements, including settlement, shrinkage, elastic shortening, floor beam deflections, creep, wind sway, twisting and racking and thermal and moisture movement.
- c. Changes due to movement of any joint in the supporting structure or building frame.
- d. Deflection of framing members under design dead loads and live loads. Incorporate these calculations in the design.
- e. Thermal and moisture induced movements of the precast cladding. Incorporate these in the design based on maximum range of temperature and humidity.

Propose nominal, minimum and maximum joint widths accounting for tolerances and movements for the SO's acceptance.

Refer to PSD for design movements and deflections to account for.

2.1.4.2 Structural Tolerances

Accommodate construction tolerances on structure as indicated in the PSD.

2.1.5 Fire and smoke performance

The overall wall construction, including any upper lining wall must meet the following criteria.

2.1.5.1 Non-combustible materials

Do not include materials that readily support combustion or add significantly to the fire load or give off toxic fumes.

2.1.5.2 Fire and smoke stopping

Prevent the passage of smoke from floor to floor in accordance with the requirements of B(C)R and BS 476: Parts 20-24.

2.1.6 Thermal performance

Achieve a U-value for the precast concrete cladding of 1.1W/m²/°C or better.

2.1.7 Acoustic performance

The total wall construction shall achieve the following STL values against noise break-in at all joints and interfaces:

Frequency(Hz)	125	250	500	1000	2000	4000
STL (dB)	25	27	32	33	30	26

Achieve resistance to flanking and floor to floor sound transmission of STC40 unless indicated otherwise in clause 3.1.

2.1.8 Functionality

2.1.8.1 Appearance

Achieve the mandatory geometry and presented on the architectural drawings.

Achieve colours and appearances presented on the architectural drawings.

Submit sample panels to illustrate the proposed surface colour, texture and variability.

2.1.8.2 Drainage

Design and install so that drainage of rainwater is predominantly at the outer face.

Provide secondary drainage at panel joints.

Provide continuity of watertightness within internal drainage routes at all panel junctions.

Ensure that drainage provision does not permit insect entry into drainage routes.

Ensure continuity of weathertightness to adjacent roof waterproofing finishes and upstands.

Ensure disposal of rainwater from precast concrete panels into rainwater disposal system.

2.1.8.3 Infestation

Resist attack or infestation by micro-organisms, fungi, insects, reptiles, birds or bats and rodents.

Have no openings in or around the precast concrete cladding that permit entry of insects, reptiles, birds or bats and rodents.

2.1.9 Lightning protection

Meet or exceed the requirements of SS CP 33. Co-ordinate as necessary between the precast concrete cladding and the main lightning protection systems for the building and shall propose appropriate connection points. No external tapes will be accepted.

2.1.10 Durability and Maintenance

Facing mixes may be used provided they are integral to the panel and do not increase the water absorption or carbonation resistance.

Do not use carbon steel in locations subject to wetting or condensation or where this will result in corrosion; shortening their design life or compromising safety, however long-term; or, causing rust, staining; or, establishing a bimetallic cell with other metals or carbon.

Identify materials and components which under normal service conditions cannot meet the specified design life and design to allow their monitoring and replacement.

Gaskets seals and baffles are to be accessible for replacement

3. MATERIALS AND COMPONENTS

3.1 Types of Precast Cladding

Refer to the PSD for the types of precast cladding and finishes requirement for the project.

3.2 Concrete

3.2.1 General

Refer to structural specifications for requirement of concrete constituents and mix design.

3.2.2 Retarders

Submit name, source, type, dosage and justification for use to SO for review.

3.2.3 Cement

Obtain all cement from a single source for the whole of the Works.

Deliver and store all cement in sealed containers.

Do not use super-sulphated or high alumina cements.

3.2.4 Water

Obtain water from PUB.

3.2.5 Pigments

Comply with the requirements in BS EN 196.

3.2.6 Sulphates

Do not exceed Total acid soluble sulphate (SO_3) content of the concrete mix of 4% of the mass of the cement in the mix.

3.2.7 Release agents

Release agents shall be one of the following types:

- a. Cream emulsion,
- b. Neat oil with surfactant added
- c. Chemical release agent

3.2.8 Reinforcement

Refer to structural specifications on Reinforcement.

3.2.9 Grouted Dowels

Use:

- a. Grade 304 stainless steel dowels.
- b. Low shrinkage grout to fill the hole completely.

3.3 Moulds

Refer to structural specifications on Precast Concrete works. Take account of the architectural finish required. The following are probable finishes:

- a. Smooth off form
- b. Water washed
- c. Honed or polished
- d. Sandblasted
- e. Acid etched
- f. Bush hammered
- g. Form liners
- h. Brick or tiled face

Retain specialised moulds for 1 year after completion.

3.4 Stone cladding finishes

3.4.1 Stone

Refer to section A1-60 clause 3.3 for selection and testing of stone. Refer to clause 3.1 for selected stone.

Determine stone thickness and submit calculations

Determine stone panel layout and joint widths to meet architectural design intent and tolerances on stone.

3.4.2 Stone fixings

Refer to section A1-60 clause 3.6.

Use stainless steel sprung clips or raked pins.

Design size and frequency and submit calculations.

Design is to allow removal and replacement of individual stones if damaged

3.4.3 Sleeves

Use Polychloroprene sleeves around pins and clips in stone.

3.4.4 Isolation layers

Use 3mm close cell polyethylene sheet.

Tape around all pins penetrating mortar.

Mortar for jointing and pointing in accordance with BS 5628-3.

3.5 Back up walls

Back up walls where indicated in the drawings may be in situ concrete, blockwork, brickwork, or steel stud and panelling. If the cavity to the wall is ventilated to the outside, the backing wall shall be designed for 100% of the wind pressure.

Do not attach precast panel restraints or fixings to non-structural back up walls, unless specific support capacity has been incorporated.

Ventilate the cavity between precast panel and back up wall

Design and install back up wall to achieve airflow not more than 1.5 cu m/hr/m² at 600 Pa. There shall be no region of concentrated airflow through the precast concrete cladding joints or at any interfaces.

Ensure that all flashings, fixings and other items within the cavity shed water outwards. Minimum spacing of flashings in the cavity shall be 10mm.

3.6 Insulation

Thermal insulation shall (be / have):

- a. Inert, durable, rot and vermin proof, CFC and HCFC free.
- b. Not support mould fungal or bacteria growth.

Allow for reduced performance of insulation due to the effects of moisture and ageing.

3.7 Waterproofing

3.7.1 Membranes

Membranes within the precast concrete cladding shall (be / have):

- a. Butyl or EPDM
- b. Mechanically fixed
- c. Formed into shapes compatible with the building interfaces
- d. Achieve necessary continuity by vulcanising the seams.
- e. Seal all joints and penetrations.
- f. UV resistant
- g. The necessary tensile strength and elasticity for their application
- h. Water vapour permeability appropriate for their application.

3.7.2 Gaskets

All gaskets shall (be / have):

- a. Extruded EPDM complying with the provisions of BS 4255.
- b. Cellular profiled gaskets shall comply with ASTM C509.

3.7.3 Sealants

Select and install in accordance with BS 6213.

Use in accordance with the manufacturer's directions, particularly relating to the use of primers.

Sealants shall (be / have):

- a. Non-gassing polyethylene closed cell foam backing rods

- b. Primers and joint preparation materials as recommended by the sealant manufacturer.
- c. Silicone sealants shall comply with BS 5889.

Refer colour selection of any visibly exposed sealants to the SO.

3.7.4 Baffles

Use polychloroprene baffles, hung from stainless steel pins and draining sequentially outwards.

Size baffles to accommodate joint tolerances and building movements.

Refer to SO for colour.

3.7.5 Membrane Flashings and Trays

Use membranes in accordance with Table 2 of BS 8297, types are:

- a. Flexible chlorosulphonated polyethylene.
- b. Flexible chloroprene. Non-cellular chloroprene products complying with the requirements of BS 4255-1.
- c. Flexible butyl rubber.

3.7.6 Metal Flashings, Trays and Cappings

Use lead, stainless steel or coated aluminium Flashing and trays

Undersleeve cappings with continuous membrane

Secure all metal flashings to precast , trays and cappings to resist full wind load

Close off cappings to prevent air passage beneath capping from elevation face to roof face and from outside to inside the building.

Flashings to be:

- a. Not less than 1.6mm thick for hidden flashings
- b. Not less than 3mm thick for components exposed to view or to impact

Isolate all galvanised steel and aluminium components in direct contact with, but not totally encased in, cementitious surfaces with thin dense PVC or EPDM isolation packs. The use of bituminous paint shall not be accepted.

3.8 Stainless Steel Components

Stainless steel components shall be austenitic, complying with BS En 10088 grades:

- a. X2CrNi18-10 (formerly 304 S11) for concealed locations only
- b. X8CrNi18-9 (formerly 304 S31) for concealed locations only
- c. X2CrNiMo17-12-2 (formerly 316 S11) for exposed locations only
- d. X2CrNiMo18-14-3 (formerly 316 S13) for exposed locations only

3.9 Carbon Steel Components

3.9.1 General

Comply with structural specifications for requirements relating to secondary steel framing.

The use of carbon steel shall only be permitted where the component is readily accessible for regular inspection and maintenance.

Where used, it shall be one of the following:

- a. Hot rolled steel complying with the requirements of BS EN 10025.
- b. Hot rolled sections to BS 4: Part 1, BS EN 10210-2 and BS EN 10021-1, tubes to BS 6323: Parts 1, 2 & 3 and angles to BS 4848: Part 4.
- c. Cold formed pre-galvanised steel sections to BS EN 10142, BS EN 10143, BS EN 10147
- d. Steel sheet to BS 1449: Part 1.

3.9.2 Steelwork Corrosion and Fire Protection and Paint Finishes

3.9.2.1 Comply with the requirements in structural specifications on Protection for structural steelwork.

3.9.2.2 Refer colours and gloss levels of any exposed items to the SO.

3.9.2.3 Finished surfaces shall be

- a. Dense and consistent
- b. Free from flow lines, streaks, pin holes, blisters, tears, lumps and other coating defects or surface imperfections when viewed from a distance of 1 metre under normal diffused daylight.

3.9.2.4 Hot dip galvanising

Hot dip galvanise all carbon steel components after forming and drilling in accordance with BS EN 1461.

Isolate all galvanised steel components in direct contact with, but not totally encased in, cementitious surfaces with thin dense PVC or EPDM isolation packs. The use of bituminous paint shall not be accepted.

3.10 Fire Stopping and Smoke Flashings

Fire stops shall be based upon mineral fibre of appropriate density, held captive by metal trays mechanically fixed to the underside of the slab and allowing for movement. Refer to clause 3.1 for the Fire resistance of the fire stops.

3.11 Fixings and Fasteners

3.11.1 General

Use stainless steel fixings and fasteners to BS EN ISO 3506 wherever wetting may occur.

Use carbon steel bolts with carbon steel components in dry locations only.

Ordinary bolts and nuts to BS EN 24032, BS EN 24033, BS EN 24034. All plain and tapered washers to BS 4320, Form E.

Nuts to BS 4190 for grade 8.8.

The use of zinc plating to BS 1706 – Fe/Zn 8 is acceptable where serrated fixings are used.

3.11.2 Brackets

Use stainless steel brackets and stainless steel fixings wherever wetting may occur (including condensation)

Brackets shall (have / be):

- a. Resist all loads, movements and dimensional changes that may occur in the building due to thermal changes, deflections, settlement and creep.
- b. Resist all loads, movements and dimensional changes that may occur to the precast concrete cladding.

Allow for construction tolerance in the all components and structure.

Adjust by small increments in and out, up and down and side to side

Use lock nuts to prevent loosening due to movements and/or vibrations.

State the torque values for tightening all bolts on the drawings submissions.

State the maximum allowable shim dimension on the drawings submissions. This shall not be greater than 10mm.

3.11.3 Cast-in Anchors

3.11.3.1 To secure precast concrete panels to building structure use Cast in anchors unless otherwise agreed with SO.

3.11.3.2 Cast in anchors incorporated in panels to receive window anchors or support bracketry for signage, lighting, rainscreens etc.

Use stainless steel wherever wetting may occur.

Attach to reinforcement to safely transmit load into precast panel

Allow for resulting dead and applied loads in precast panel and fixing bracket, anchor design.

3.12 Aluminium Windows

3.12.1 Cast-in Windows

Protect windows from dirt, grout, Provide adequate lapping over the joints of two protective tapes and other deleterious material by wrapping with self adhesive protective tape.

3.12.2 Site Installed Windows

Ensure that the window fits into the opening. No hacking of the panel will be allowed.

4. WORKMANSHIP

4.1 Design and Selection of Architectural Quality of Finish

Provide Samples for SO review and selection of finish.

When finish is selected produce 1m² finish range samples for review by SO.

Agree 5 accepted 1m² finish range samples with SO.

Saw these into three equal pieces retain one set at the factory, one at site and one with the SO.

Identify concrete mix and sources, mould properties casting and curing, post casting treatments and other details to allow consistent panels to be produced.

Facing mixes may be used provided they are integral to the panel and do not increase the water absorption or carbonation resistance.

4.2 Fabrication

4.2.1 Factory Procedures

Submit proposals for materials delivery, storage and batching

Identify every panel uniquely so that it is traceable through materials used, cube results, stage inspection records.

Do not give a replacement unit the same reference as the rejected unit it replaces.

4.2.2 Secondary Steel Framework

For workmanship requirement refer to structural specifications on Structural Steelwork.

4.2.3 Applying Release Agents

Use release agents in accordance with their manufacturer's recommendations.

4.2.4 Placing Stone Finishes in Moulds

Check edges of stones so that polished faces are outermost.

Place stones, spacers and sand in joints, isolation layer steel clips or pins and rubber grommets.

Identify any non-conformances with proposed remedial measures or rejection.

Submit record of inspection to the SO.

4.2.5 Concreting

For assembling reinforcement cages, casting, curing, and striking, refer to structural specifications on precast concrete works.

4.2.6 Finishes

Remove spacers and surface contaminants / debris trapped in face.

Apply any post applied surface treatments required to achieve the architectural quality finish.

4.2.6.1 Architectural quality concrete panel finishes

Consistently achieve the architectural quality concrete panel finishes as determined by samples submitted for review to the SO.

Facing mixes may be used but these must be integral with the backing mix ie no voids or de-bonding and provide low permeability cover to reinforcement such that water absorption and carbonation resistance are not compromised.

Identify any post casting treatments used to achieve the required appearance

Submit details of mix designs, cement and aggregate sources, mould face design and materials, casting and curing procedure and any other details required to accurately replicate the finish.

Unless otherwise noted in clause 3.1, unformed surfaces (face up casting) to be steel trowelled finish.

4.2.6.2 Smooth off form

AS 3610 sets out five categories of off form finish. Class1 is exceptional and Class 5 concealed. Refer to clause 3.1 for the required level of finish.

The texture of the surface is described on the drawings. Allow for the preparation of three sample panels 1.2mx1.2m to agree on the extent colour and texture of the off form surface.

Do not undertake any repairs without SO approval.

4.2.6.3 Water washed

Water wash the stone to expose one third of the aggregate, two thirds should remain embedded. The colour, shape and size of the aggregate is set out in clause 3.1.

Agree with the SO, the extent of retarder used to control the removal of the matrix.

On completion of the water washing, wash the surface with 5% muriatic acid, wash with clean water to remove all trace of the acid.

Allow for the preparation of three sample panels 1.2mx1.2m to agree on the extent colour and texture of water washed surface.

4.2.6.4 Honed or polished

Hone and polish the concrete panels with Grit size as noted in clause 3.1 to achieve the polished level required. Cast panels face down to achieve flat a surface.

Allow for the preparation of three sample panels 1.2mx1.2m to agree on the extent and level of acceptable polish.

4.2.6.5 Sandblasted

Sand or abrasive blast panel faces to achieve the surface finish described on the drawings.

Allow for the preparation of three sample panels 1.2mx1.2m to agree on the extent and depth of the sand/abrasive.

Where noted on the drawings mask off areas of the panels to protect the surface from blasting.

Wash with weak-5% solution of hydrochloric acid after sand blasting, wash off all acid with clean water.

4.2.6.6 Acid etched

Acid etch surfaces two weeks after casting using low strength solutions of acid. Flush with clean water 15 minutes after application of the acid.

Ensure consistency of the acid etching process –washing solution, strength, timing and protect all metal parts with chlorinated rubber paint prior to the application of the acid.

Set out remedial work for the repair of any galvanised elements damaged by the acid. Submit to SO for approval prior undertaking the repair.

Allow for the preparation of three sample panels 1.2mx1.2m to agree on the extent of the acid etching.

4.2.6.7 Bush hammered

Scabble or hammer off face from concrete surface after formation. Achieve a uniform surface. Produce shop drawings setting out form of the concrete surface with an indication of the extent of the scabbling/hammering.

Allow for the preparation of three sample panels 1.2mx1.2m to agree on the extent of the scabbling.

4.2.6.8 Form liners

Insert Silicone or polyurethane liners into the mould to form the pattern set out in the drawings. Refer to clause 3.1 for the concrete mix if any.

Allow for the preparation of three sample panels 1.2mx1.2m to agree on the standard of form liner finish.

4.2.6.9 Brick or tile face

Set out bricks and tiles on rubber mats on the bottom of the mould. Refer to clause 3.1 for the brick/tile and joint specification.

Allow for the preparation of three sample panels 1.2mx1.2m to agree on the standard of tile/brick finish.

4.2.7 Factory Finishing Stone Facings

Clean out joints of sand from casting or grout leaks.

Inspect for alignment of stones and joints and damage:

- a. If correct grout or seal joints to achieve architectural design intent.
- b. If incorrect propose and carry out remedial measures.

4.2.8 Factory Installation of Windows

Refer to section A1-20 "Windows" clause 4.0.

4.2.9 Factory Sealant Work

Apply and cure sealants in accordance with their manufacturer's directions on minimum and maximum temperature and RH conditions.

Achieve the depth to width geometry.

Prevent three-sided adhesion.

Use appropriate backer rods, tapes, surface preparation and primers as manufacturer's directions.

Tool off sealants neatly.

4.2.10 Fabrication of Metal Components

Mechanically cut and drill metals wherever possible. Cut surfaces true to line and level and free from burrs. Cut sections that form the contact surfaces of any joint free of irregularities and unevenness and dress for welding as necessary.

Fabricate all cappings and flashings on a brake press.

4.2.11 Inspection at Factory

Comply with section 9.2 of BS 8297

Check for any honeycombing / voids close to the surface, check cover, finish quality and visual consistency.

Inspect for consistency of finish and compliance with agreed range samples.

Identify any non-conformances with proposed remedial measures or rejection.

Identify cause and implement measures at source to prevent recurrence.

Submit record of inspection to the SO.

4.2.12 Factory Repairs

4.2.12.1 Damaged panels

Reject all damaged panels unless remedial measures were accepted and carried out to SO's satisfaction.

4.2.12.2 Repair concession

In the event of a concession being given for the repair:

- a. Submit a proposed repair procedure and carry out trials.
- b. Repairs on surfaces visible in the installed works shall not be visually discernable both in a dry condition and a wet condition. Repairs shall weather in the same manner as the adjacent panel surface.
- c. Do not repair in prominent locations
- d. Do not compromise the overall performance of the pre-cast panels and their fixings.

4.2.12.3 Cosmetic repairs

Subject to authorisation by the SO and review of the results, cosmetic repairs to surface chips, blowholes less than 20mm in area and 6mm depth may be permitted.

4.2.12.4 Damage exposing reinforcement

Subject to authorisation by the SO and review of the results, damage exposing reinforcement may be repaired provided the repair addresses protection of reinforcement, priming of exposed and damaged edges, appropriate concrete mix or epoxy resin to achieve compaction, carbonation resistance and matched appearance where visible. Submit details of materials method, attachment of shuttering and striking.

4.2.12.5 Voids and honeycombing

Subject to authorisation by the SO and review of the results, localised voids and honeycombing may be repaired. Determine the cause and address at source in order to prevent recurrence. Cut back to sound concrete and proceed as for damage exposing reinforcement procedure. If voids and honeycombing are widespread, reject the panel.

4.2.12.6 Cracks

Subject to authorisation by the SO and review of the results, cosmetic repairs to localised shrinkage cracks may be repaired. Proceed as for cosmetic repair procedure.

Reject panels with cracks that pass through the body of the panel.

4.2.12.7 Structural damage

Reject panels with damage affecting their structural behaviour.

Inspect repaired panels and identify any non-conformances in repair procedures along with proposed remedial measures or rejection.

Submit record of inspection to the SO who may review the repair and accept or reject the panel.

Complete all repairs before panels leave the factory.

4.3 Fabrication and Installation - Handling

Protect and handle units to prevent impact damage, cracking or distortion.

Handle precast cladding units by designed strong points only.

4.4 Pre-cast Concrete Panel Installation

4.4.1 General

Install in accordance with BS8297, section 11 and the Architectural Cladding Association "Code of Practice for Safe Erection of Pre-cast Concrete Cladding".

4.4.2 Line and Level Survey

Undertake a survey of line and level and condition of structure to verify its acceptability to receive precast concrete cladding.

4.4.3 Site Setting Out Marks

Use removeable marking or place marks in locations where they will be concealed in the Works when installed.

4.4.4 Tolerances

Achieve tolerance of erection as specified in BS 8297, section 11.5.

4.4.5 Cast-in Anchors

Clean out all cast-in anchors.

Verify correct position and secure connection to building structure by inspection of all units and test 2% per floor.

4.4.6 Fixings

Install fixings in accordance with the requirements of clauses 11.3.4 and 11.3.5 of BS8297 to correct torque and in accordance with their manufacturer's instructions and procedures.

Ensure the strength grade combination of bolts, nuts and washers as specified.

Use a torque spanner to achieve correct tightening of fixings to achieve correct tension load and avoid under / overstressing,

Use lock-nuts to prevent loosening.

At least one clear thread shall show above the nut.

Ensure allowance for movement where required at fixings and dowels.

4.4.7 Bedding, Jointing and Sealing

Carry out bedding, jointing and sealing in accordance with clause 11.4 of BS8297.

4.4.8 Grouting of Corbels

Grout between corbel and structural slab with non-shrink, flowable high strength grout.

4.4.9 Capping Lifting Ferrules

Cap lifting ferrules etc. after installation so that they cannot fill with water and cause rust staining or persistent wet patches.

Where exposed to direct sunlight use mortar

Where concealed from direct sunlight plastic caps may be used.

4.4.10 Fire Protection of Brackets and Fixings

Ensure 2-hours fire protection to brackets and fixings that support the dead load of the precast panels.

4.4.11 Installation of Insulation

Fix insulation on pins or by wires so that it will not slump.

Fill all gaps and discontinuities.

4.4.12 Installation of Vapour Barriers

Seal and clamp vapour barriers to achieve continuity around fixings laps etc.

4.4.13 Installation of Baffles and Gaskets

Install baffles and gaskets without incorrect distortion such as stretching or compression of length, or folding back of wiper seals.

4.5 Site Sealant Work

Apply and cure sealants in accordance with their manufacturer's directions on minimum and maximum temperature and RH conditions.

Achieve the depth to width geometry.

Prevent three-sided adhesion.

Use appropriate backer rods, tapes, surface preparation and primers as manufacturer's directions.

Tool off sealants neatly.

4.6 Installation of Fire and Smoke Stopping

Install fire and smoke stopping at all voids and penetrations between compartments at designated locations to achieve the specified fire and smoke stopping performance.

4.7 Site Repairs

Inspect immediately after at installation for damage.

Repair cosmetic damage and subject to review by SO.

Repair impact damage and subject to review by SO.

Remove any cracked or structurally damaged panels and replace with new panels whilst panel can still be removed.

In the event of damage at a later stage when a panel cannot be taken out, submit proposals to the SO for review.

4.8 Siloxane Treatment

Refer to clause 3.1 as to whether Siloxane is to be applied to the cladding.

Clean all mould or retarder oil from surface before applying Siloxane.

4.9 Protection and Cleaning

Protect vulnerable surfaces, edges, corners and features. Apply temporary removable protection against damage from following trades or operations and the anticipated hazards. Materials used for temporary protection shall be compatible with and subsequently removed from the surface and finishes without detriment to the finish. The component suppliers and finishers shall approve methods for removal.

Unless agreed otherwise by the SO protection should remain in place until all work potentially damaging to the components or their finish has been undertaken.

Replace any damaged components in the precast concrete cladding. Concession to this requirement will only be given if the proposed repair procedure is authorised by the SO and complete and warranted repair of the damage is achieved.

5. VERIFICATION AND SUBMISSION

5.1 Submissions

Read in conjunction with submission requirement of structural specifications on precast concrete works.

5.1.1 Tender Submissions

Submit the following during tender:

- a. Material descriptions and sources of concrete mixes, gaskets, seals, linings, finishes, fixings and fasteners, vapour barriers, insulation fire stopping and barriers
- b. Junctions of principle components
- c. Position of both weather and air seals
- d. Vapour barrier and seals
- e. Methods of ventilation and drainage of the assembly
- f. Warranty periods for the principal components together with any limitations on these
- g. Drawings illustrating proposed waterproofing and drainage detail at vertical, horizontal and intersection panel to panel joints; and proposed interfaces to windows and adjacent cladding
- h. 1.000m x 1.000m sample of finish for each facing type.

5.1.2 Work Submissions

5.1.2.1 Method statements describing:

- a. Proposals to fabricate and install panels
- b. Concrete samples and mix designs
- c. Description of all moulds.
- d. Criteria for re-use of moulds with limits identified for wear, damage, distortion

Refer to section A1-60, clause 4.1 for requirements on stone selection procedures.

5.1.2.2 Detail design drawings

Mix designs, cement and aggregate sources, admixtures, mould face design agents and materials, casting and curing procedure and any other details required to accurately replicate the finish.

Stone thickness calculations.

Data and calculations to verify that structural, acoustic, condensation analysis, thermal transmittance and weathering, can be met for each panel.

5.1.3 Fabrication Submissions

- a. Mould inspection records
- b. Stones in moulds records

- c. Damage reports and repair proposals

5.1.4 Installation

- a. Proposed locations for hose test.
- b. Site hose test results and certificates.
- c. Proposals to address any shortcomings identified in hose testing.
- d. Site hose test reports and remedial measures.

5.1.5 Maintenance Submissions

Include the following information in the maintenance manual and logbook:

- a. Cleaning methods for panel surfaces.

5.1.6 Warranty

Submit the warranty for the SO's acceptance upon completion of the works.

5.1.7 Quality Control Plan Submission

Prepare and submit the quality control plan for SO's acceptance prior to starting work.

5.2 Samples and Mock-ups

5.2.1 Samples

Submit samples, prototypes and demonstrations of fabrication and installation as required by SO.

Following review of samples, prototypes and demonstrations all subsequent materials and workmanship are to be the same or better standard.

Four sample panels to illustrate the major junctions of the elevation.

5.2.2 Mock-up

Refer to the PSD for requirement of mock-up panels for the project, if any.

Undertake visual mockup.

5.3 Inspection

SO to inspect all pre cast panels before they are delivered to site.

5.4 Tests

5.4.1 Water Tightness Test to External Full Height Precast Walls

5.4.1.1 General

Carry out tests on 10% of the external walls directly facing the exterior. Excluding access balcony walls, parapet walls or determined by the SO to be excluded.

5.4.1.2 Source of water

Water used to test the water tank prior to its sterilisation. Sterilise the tank after completion of the water test.

Use PUB potable water only for re-test.

5.4.1.3 Method of testing

Provide the following information to the SO for approval eight weeks before carrying out the test:

- a. The equipment set up to conduct the test
- b. Procedure of the water test
- c. Pump capacity to deliver the required flow rate
- d. Method to suspend the nozzle

Fix nozzle 1800mm-2000mm away from the surface of the wall, at an incline of 30° to the external wall as shown in the drawings.

Deliver 300 litres of water for 2 hours.

Place the nozzle so that it covers the whole wall panel.

The panel will be considered to have passed the test if no dampness or seepage appears on the internal surface of the wall panel or adjacent areas during the spraying and within half an hour of completion of the spraying.

5.4.1.4 Further testing

Should more than 20% of the panels fail the test a further 10% of the panels are to be tested. Should more than 20% of these panels fail all other panels are to be tested.